Supplementary Information

Out of Time, Out of Rights?

Limitation Extensions and Access to Justice in Employment Tribunals

Supplementary methods, coding documentation, validation and reproducibility information

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S1. Search Strategy and Sampling

S1.1 Data source and eligibility period

The data comprised published first-instance Employment Tribunal decisions available through the official GOV.UK Employment Tribunal decisions database. The database was searched on Apri 23rd 2026. Decisions were eligible if they were decided between 1 January 2025 and 31 December 2025 and contained a substantive limitation issue concerning unfair dismissal, discrimination, or both. Employment Appeal Tribunal and other appellate decisions were excluded from the coded empirical sample.

S1.2 Search terms and filters

Two search streams were used.

1. **Unfair-dismissal stream.** The "Unfair Dismissal" jurisdiction filter was combined with the keywords "out of time" and "reasonably practicable".
2. **Discrimination stream.** The keywords "out of time" and "just and equitable" were combined with the discrimination jurisdiction filters "Age Discrimination", "Disability Discrimination", "Race Discrimination", "Religion or Belief Discrimination", "Sex Discrimination", "Sexual Orientation Discrimination/Transexualism", "Maternity and Pregnancy Rights" and "Victimisation Discrimination".

Other claims did not make a decision ineligible where unfair dismissal, discrimination, or both were also present. Judgments involving both principal claim types could be retained, but duplicate decisions were counted only once.

S1.3 Inclusion, exclusion and screening

A single legal reviewer screened results sequentially within each search stream until 75 eligible decisions had been identified. Decisions were included when the published reasons addressed whether an unfair-dismissal or discrimination claim was in time, whether time should be extended, or the application of the "not reasonably practicable" or "just and equitable" test.

The following were excluded:

- duplicate records or duplicate judgments;
- decisions concerned only with withdrawal or strike-out and containing no substantive limitation reasoning;
- judgments without reasons sufficient for structured coding; and
- appellate judgments.

S2. Coding Framework

S2.1 Development and pilot testing

A structured codebook was developed from the research questions and an initial review of Employment Tribunal judgments. The preliminary framework was piloted on 10 judgments covering unfair dismissal and discrimination. The pilot was used to refine variable names, permitted categorical values, claim-specific outcomes, reasoning-factor labels, missing-data conventions and instructions for recording supporting evidence.

S2.2 Definition of "materially weighed"

For the final reasoning-factor analysis, a factor was counted only where the tribunal materially used it in deciding whether to grant or refuse an extension. A factor could count on either side of reasoning: it could support an extension, or its evaluation and rejection could materially support refusal.

S2.3 Model and settings

Each judgment PDF was processed separately through a purpose-built R pipeline. The extraction request was sent to the Anthropic Messages API using the following settings:

- model identifier: `claude-sonnet-4-5`;
- temperature: 0;
- maximum output tokens: 5,000

S2.4 Complete extraction prompt

System prompt

```
You are a rigorous legal content analysis assistant.
Apply the recurrent rules below to all coding tasks and maintain deterministic behaviour.

1) Code only information supported by the Employment Tribunal decision text.
Do not invent missing facts, dates, claims, outcomes, reasons, or legal findings.

2) Record uncertainty explicitly. If information is not stated or cannot be determined,
use "unclear", "not_applicable", or null as appropriate.

3) Distinguish unfair dismissal from discrimination claims. For discrimination claims,
identify whether the "just and equitable" test is applied or discussed; for unfair
dismissal claims, identify whether the "reasonably practicable" test is applied or discussed.
If both appear, code extension_test as "both".

4) If only part of a claim or some allegations are out of time, code the relevant out_of_time field
as "partly" if available in the codebook, otherwise code as "yes", and explain the qualification
in the relevant reasoning field.

5) Return valid JSON only. Include "_model", "_run_id", and "case_id";
use no trailing commas. If confidence is low, explain why in "coding_notes".
```

User prompt template

```
Analyse the Employment Tribunal decision according to the predefined variables in the codebook.
Make only claims backed by the decision text. Do not alter the variable names or permitted categorical values
     .

INPUTS
- Decision text: <DECISION_TEXT>
- Codebook: <CODEBOOK_JSON>

OUTPUT
Return one valid JSON object only. Do not include markdown, prose, or code fences.

Use this metadata exactly:
"_model": "<MODEL_NAME>"
"_run_id": "<RUN_ID>"
"case_id": "<CASE_ID>"

The JSON object must include every top-level field in the supplied codebook.
It must also include evidence and coding_notes if those fields are included in the codebook.

CONSTRAINTS
- Global recurrent coding rules apply.
- Do not treat a limitation issue as present merely because "time limits" appears in a list of claims.
- The decision must discuss whether the claim or allegation was presented in time, or whether time should be
     extended.
- If the claim is partly in time and partly out of time, record the nuance in the relevant reasoning field.
- Use only the permitted categorical values in the codebook.
- Use null where a numeric or date value cannot be determined.
```

```
CODEBOOK_JSON:
<CODEBOOK_JSON>

DECISION_TEXT:
<DECISION_TEXT>
```

S2.5 Extraction JSON codebook

Listing 1: Complete JSON extraction codebook

```
{
  "case_name": {
    "type": "string",
    "nullable": false,
    "description": "Name of the case as stated in the tribunal decision."
  },
  "tribunal_region": {
    "type": "string",
    "nullable": true,
    "description": "Tribunal region or hearing location, if stated in the decision."
  },
  "decision_date": {
    "type": "date",
    "format": "YYYY-MM-DD",
    "nullable": true,
    "description": "Date of the tribunal decision in ISO format."
  },
  "type_of_claim": {
    "type": "categorical",
    "values": [
      "unfair_dismissal",
      "discrimination",
      "both",
      "neither"
    ],
    "nullable": false,
    "description": "Primary claim type relevant to the study."
  },
  "secondary_claims": {
    "type": "string",
    "nullable": true,
    "description": "Other claims mentioned in the decision, such as wrongful dismissal, holiday pay, wages,
     redundancy pay, victimisation, harassment, or breach of contract."
  },
  "discrimination_type": {
    "type": "categorical_or_array",
    "values": [
      "sex",
      "race",
      "disability",
      "age",
      "religion_belief",
      "sexual_orientation",
      "pregnancy_maternity",
      "marriage_civil_partnership",
      "gender_reassignment",
      "other",
      "not_applicable",
      "unclear"
    ],
    "nullable": false,
    "description": "Protected characteristic or discrimination category involved in the claim. Use an array
     if more than one discrimination type is clearly present."
  },
  "claim_outcome_unfair_dismissal": {
    "type": "string",
    "nullable": true,
    "description": "Outcome of the unfair dismissal claim, if present."
```

```
},
"claim_outcome_discrimination": {
  "type": "string",
  "nullable": true,
  "description": "Outcome of the discrimination claim, if present."
},
"out_of_time_unfair_dismissal": {
  "type": "categorical",
  "values": [
    "yes",
    "no",
    "partly",
    "unclear",
    "not_applicable"
  ],
  "nullable": true,
  "description": "Whether the unfair dismissal claim, or any part of it, was found to be out of time."
},
"out_of_time_discrimination": {
  "type": "categorical",
  "values": [
    "yes",
    "no",
    "partly",
    "unclear",
    "not_applicable"
  ],
  "nullable": true,
  "description": "Whether the discrimination claim, or any part of it, was found to be out of time."
},
"reason_for_out_of_time_unfair_dismissal": {
  "type": "string",
  "nullable": true,
  "description": "Tribunal's explanation of why the unfair dismissal claim was found to be out of time."
},
"reason_for_out_of_time_discrimination": {
  "type": "string",
  "nullable": true,
  "description": "Tribunal's explanation of why the discrimination claim was found to be out of time."
},
"delay_length_days": {
  "type": "number",
  "nullable": true,
  "description": "Number of days by which the relevant claim was late, if calculable from the decision.
   Where unfair dismissal and discrimination have different delay periods, explain this in coding_notes."
},
"delay_reason": {
  "type": "string",
  "nullable": true,
  "description": "Reason given by the claimant, tribunal, or evidence for the delay. Where reasons differ
   by claim type, explain this in the claim-specific reasoning fields."
},
"extension_requested_unfair_dismissal": {
  "type": "categorical",
  "values": [
    "yes",
    "no",
    "implicit",
    "unclear",
    "not_applicable"
  ],
  "nullable": true,
  "description": "Whether the claimant requested, relied on, or was treated as seeking an extension of time
    for the unfair dismissal claim."
},
"extension_requested_discrimination": {
  "type": "categorical",
  "values": [
    "yes",
    "no",
    "implicit",
```

```
      "unclear",
      "not_applicable"
    ],
    "nullable": true,
    "description": "Whether the claimant requested, relied on, or was treated as seeking an extension of time
     for the discrimination claim."
  },
  "extension_test": {
    "type": "categorical",
    "values": [
      "just_and_equitable",
      "reasonably_practicable",
      "both",
      "other",
      "not_applicable",
      "unclear"
    ],
    "nullable": true,
    "description": "Legal test or tests applied or discussed when considering whether to extend time."
  },
  "extension_outcome_unfair_dismissal": {
    "type": "categorical",
    "values": [
      "granted",
      "refused",
      "partially_granted",
      "not_decided",
      "other",
      "not_applicable",
      "unclear"
    ],
    "nullable": true,
    "description": "Outcome of the extension issue for the unfair dismissal claim."
  },
  "extension_outcome_discrimination": {
    "type": "categorical",
    "values": [
      "granted",
      "refused",
      "partially_granted",
      "not_decided",
      "other",
      "not_applicable",
      "unclear"
    ],
    "nullable": true,
    "description": "Outcome of the extension issue for the discrimination claim."
  },
  "reasoning_unfair_dismissal": {
    "type": "string",
    "nullable": true,
    "description": "Brief summary of the tribunal's reasoning on limitation, time limits, and/or extension
     for the unfair dismissal claim."
  },
  "reasoning_discrimination": {
    "type": "string",
    "nullable": true,
    "description": "Brief summary of the tribunal's reasoning on limitation, time limits, and/or extension
     for the discrimination claim."
  },
  "reasoning_factors_unfair_dismissal": {
    "type": "array",
    "items": "string",
    "nullable": true,
    "allowed_examples": [
      "length_of_delay",
      "prejudice",
      "illness",
      "disability",
      "mental_health",
      "lack_of_advice",
```

```
      "legal_advice_error",
      "internal_grievance",
      "internal_appeal",
      "acas_early_conciliation",
      "ignorance_of_rights",
      "promptness",
      "effective_date_of_termination",
      "technical_or_filing_problem",
      "merits",
      "other"
    ],
    "description": "Factors relied on by the tribunal when reasoning about limitation or extension for the
     unfair dismissal claim."
  },
  "reasoning_factors_discrimination": {
    "type": "array",
    "items": "string",
    "nullable": true,
    "allowed_examples": [
      "length_of_delay",
      "prejudice",
      "illness",
      "disability",
      "mental_health",
      "lack_of_advice",
      "legal_advice_error",
      "internal_grievance",
      "internal_appeal",
      "acas_early_conciliation",
      "ignorance_of_rights",
      "promptness",
      "continuing_act",
      "conduct_extending_over_period",
      "merits",
      "other"
    ],
    "description": "Factors relied on by the tribunal when reasoning about limitation or extension for the
     discrimination claim."
  },
  "claimant_representation": {
    "type": "categorical",
    "values": [
      "represented",
      "unrepresented",
      "partially_represented",
      "did_not_attend",
      "unclear"
    ],
    "nullable": true,
    "description": "Whether the claimant was represented, unrepresented, partially represented, did not
     attend, or unclear from the decision."
  },
  "respondent_representation": {
    "type": "categorical",
    "values": [
      "represented",
      "unrepresented",
      "partially_represented",
      "did_not_attend",
      "unclear"
    ],
    "nullable": true,
    "description": "Whether the respondent was represented, unrepresented, partially represented, did not
     attend, or unclear from the decision."
  },
  "barriers_to_access": {
    "type": "string",
    "nullable": true,
    "description": "Any comments made by the tribunal, or cautiously inferable from the decision, about
     barriers to accessing the tribunal process."
  },
```

```
  "ai_confidence": {
    "type": "categorical",
    "values": [
      "high",
      "medium",
      "low"
    ],
    "nullable": false,
    "description": "Confidence level assigned by the AI coder for this coding decision."
  },
  "coding_notes": {
    "type": "string",
    "nullable": true,
    "description": "Notes for human reviewer, especially where confidence is low, the decision is ambiguous,
     or the output requires manual checking."
  }
}
```

S3. Human Validation

S3.1 Validation sample and manual review

Sixteen judgments (10.7% of the 150-case dataset) were randomly selected using a random-number generator in R. A single legal researcher independently reviewed the original judgments, blinded to the LLM-generated classifications, and manually coded the principal variables and reasoning factors. Human classifications were subsequently compared with the corresponding LLM outputs.

S3.2 Harmonisation and agreement calculations

Equivalent categorical labels were standardised before comparison. Missing and non-applicable observations were handled according to variable-specific eligibility criteria, resulting in different denominators across comparisons. Reasoning factors were assessed as binary indicators of presence or absence within applicable claim-specific determinations. The validation examined 17 individual factor labels, whereas the final analysis combined `continuing_act` and `conduct_extending_over_period` into one of 16 categories.

Percentage agreement was calculated as the proportion of exact agreements among eligible paired comparisons. Cohen's kappa (κ) was calculated for categorical variables to account for agreement expected by chance.

S3.3 Human–LLM agreement results

Agreement was observed in 179 of 205 principal-variable comparisons (87.3%) and 198 of 233 reasoning-factor comparisons (85.0%), giving an overall descriptive agreement of 377/438 (86.1%). Claim-type agreement was 75.0% (12/16; $\kappa = 0.64$). Agreement varied between factors, with lower agreement for apparent merits (60.0%; $\kappa = 0.06$) and effective date of termination (57.1%; $\kappa = 0.22$).

The validation sample was relatively small, and individual judgments contributed multiple comparisons. Consequently, the pooled agreement percentage is descriptive and does not establish the accuracy of all LLM-generated classifications.

S4. Statistical Methods

S4.1 Descriptive analyses and denominators

Analyses were conducted in R version 4.5.2. Categorical variables were summarised as counts and percentages. Continuous delay length was summarised using the median, interquartile range and range because its distribution was strongly right-skewed.

Denominators were analysis-specific:

- overall case characteristics used all 150 unique coded judgments unless otherwise stated;
- discrimination-ground percentages used cases containing a discrimination claim;
- out-of-time findings used claim-applicable cases with a coded finding;
- extension outcomes used claim-applicable cases with a coded outcome;
- case-level association analyses used unique cases with at least one decided extension outcome; and
- the final materially-weighted factor table used all claim-applicable cases (106 unfair-dismissal and 111 discrimination cases), including cases in which a decided extension application was not recorded.

Cases involving both claim types could contribute to both claim-specific denominators. Reasoning-factor categories were non-exclusive.

S4.2 Unadjusted odds ratios

For the forest plot, a case was eligible if at least one extension outcome was decided. A successful case-level outcome was defined as a granted unfair-dismissal extension or a granted or partially granted discrimination extension. For each binary characteristic, the unadjusted odds ratio was calculated from a 2×2 table:

$$\mathrm{OR} = \frac{ad}{bc},$$

where a and b were successful and unsuccessful outcomes among exposed cases, and c and d were successful and unsuccessful outcomes among unexposed cases. If any cell was zero, 0.5 was added to all four cells (Haldane-Anscombe correction). Wald 95% confidence intervals were calculated on the log scale:

$$\exp\left\{\log(\mathrm{OR}) \pm 1.96\sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}\right\}.$$

Characteristics examined were inclusion of unfair dismissal, inclusion of discrimination, claimant unrepresented status, race discrimination, sex discrimination, disability discrimination, multiple discrimination grounds and application of the "reasonably practicable" test alone. These estimates were unadjusted, no multiplicity correction was applied, and results were interpreted descriptively rather than causally.

S4.3 The 92-day proxy for a six-month primary period

The coded delay represented days beyond the existing primary deadline. Moving from an approximately three-month to a six-month primary period adds approximately three calendar months.

The analysis therefore used 92 days as an indicative proxy for that additional period. Among decided determinations with a known numerical delay, the analysis calculated the proportion with `delay_length_days <= 92`, overall and within each delay-related factor.

This was not an exact legal recalculation. It estimates the broad proportion of historically late claims that might have fallen within an additional three-month window; it does not predict how claimant behaviour would change under the reform.

S4.4 Reasoning-factor analysis

Reasoning factors were initially identified from tribunal judgments through LLM-assisted extraction and recorded in claim-specific structured fields. Subsequent analysis was conducted in R using rule-based text parsing of these extracted labels. Records with blank `type_of_claim` fields were excluded. Unfair-dismissal and discrimination claims were analysed separately, with cases involving both claims contributing to both analyses.

The R script examined 16 prespecified categories. Classification was based on exact matches between the normalised extracted labels and the predefined codes in Table S1.

Table 1: Reasoning-factor categories and matching rules

Category	Matching code and intended meaning
Length of delay	`length_of_delay`: Duration or degree of lateness evaluated by the tribunal.
Promptness	`promptness`: Whether the claimant acted promptly after an obstacle ceased or was discovered.
Lack of advice	`lack_of_advice`: Absence or availability of advice considered in the extension decision.
Ignorance of rights	`ignorance_of_rights`: Ignorance or misunderstanding of rights, time limits or procedures.
ACAS early conciliation	`acas_early_conciliation`: Timing or operation of early conciliation considered.
Effective date of termination	`effective_date_of_termination`: The termination date or misunderstanding of that date considered.
Internal appeal	`internal_appeal`: An internal appeal or its timing considered.
Internal grievance	`internal_grievance`: An internal grievance or its timing considered.
Mental health	`mental_health`: Mental health considered as an explanation for delay or its rejection.
Physical illness	`illness`: Physical illness considered as an explanation for delay or its rejection.
Disability	`disability`: Disability considered in the extension decision.
Prejudice	`prejudice`: Prejudice to either party or evidential prejudice considered.
Apparent merits	`merits`: Apparent merits or prospects of the claim considered.
Legal or representative error	`legal_advice_error`: Errors by legal advisers, unions or other representatives considered.
Technical or filing problem	`technical_or_filing_problem`: Technical, administrative or filing difficulties considered.

Category	Matching code and intended meaning
Conduct extending over a period	`conduct_extending_over_period` or `continuing_act`: Continuing conduct considered in the limitation decision.

For each applicable claim and category, a binary indicator was generated. A factor was counted as present only when (i) the relevant extension outcome was `granted`, `refused` or `partially_granted`; (ii) the corresponding tribunal reasoning field contained substantive text; and (iii) the normalised structured factor field contained the matching code. Otherwise, the indicator was zero.

Denominators included all claim-applicable cases (111 discrimination and 106 unfair dismissal), including those without decided extension outcomes, which contributed zero to factor counts. Categories were not mutually exclusive, and percentages were calculated using the respective claim-specific denominators.